

Shear and necking instabilities in nonlinear elasticity

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ABSTRACT

This paper examines the behavior of a class of nonlinearly elastic rods under tension. These rods can suffer longitudinal extension, transverse contraction, shear, and flexure. The deformation is governed by an eighth order, quasilinear system of ordinary differential equations. A convergent perturbation process shows that there is a very rich collection of solutions that bifurcate from the trivial affine configuration. Global results are obtained for hyperelastic rods. These solutions describe instabilities of the shear and necking type and can exhibit hysteresis.

RÉSUMÉ

On considère le comportement d'une classe de poutres nonlinéairement élastiques en état de tension. Ces poutres peuvent souffrir une extension longitudinale, une contraction transversale, un cisaillement et une flexion. La déformation est gouvernée par un système quasi-linéaire d'équations différentielles ordinaires du huitième ordre. Un processus convergent de perturbations montre qu'il y a une collection très riche de solutions qui bifurquent de la configuration banale affine. Dans le cas hyperélastique on obtient des résultats globaux. On montre que ces solutions décrivent des instabilités du type de cisaillement et d'étranglement et peuvent exhiber une hystérésis.

1. Introduction

In this paper we examine the behavior of a class of nonlinearly elastic rods under tension. These rods can suffer longitudinal extension, transverse contraction, shear, and flexure. The deformation is governed by a boundary value problem for an eighth-order system of quasilinear ordinary differential equations. This problem admits a family of "trivial solutions" that correspond to affine deformations. By means of a convergent perturbation process, we show that a very rich collection of nontrivial solutions can bifurcate from these trivial solutions. An examination of

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these non-trivial solution branches shows that they describe instabilities of the shear (yield-point) and necking types. We devote most of our attention to the study of the shear instability, which in real materials is accompanied by the formation of Lüders bands (cf. [10, 14]). We but briefly develop a number of new results for the technically easier necking problem, which was carefully examined in [2, 4]. We finally indicate how our model may give rise to solutions exhibiting hysteresis. In short, we demonstrate that our equations of nonlinear elasticity, endowed with neither a yield criterion nor any sort of memory, have solutions that possess many of the properties thought of as typifying the theory of plasticity.

2. Formulation of the governing equations

We study the planar equilibrium configurations of an initially straight, nonlinearly elastic, prismatic rod that can suffer longitudinal extension, transverse contraction, shear, and flexure under the action of certain loads applied at the ends. In our theory, the configuration of such a rod is defined by a pair of coplanar vector functions

$$S \in [0, 1] \mapsto \mathbf{r}(S), \mathbf{p}(S). \quad (2.1)$$

We denote differentiation with respect to S by a prime. The plane curve defined by $\mathbf{r}$ is called the *axis*. $\mathbf{p}(S)$ is called the *director* at S . (We may interpret S as representing the (scaled) arc length parameter of the line of centroids of the rod in its reference configuration, $\mathbf{r}(S)$ as representing the position of a particle on this material line of centroids in a deformed configuration, and $\mathbf{p}(S)$ as characterizing the orientation and stretch of the material cross section S of the rod. See Figure 1.)

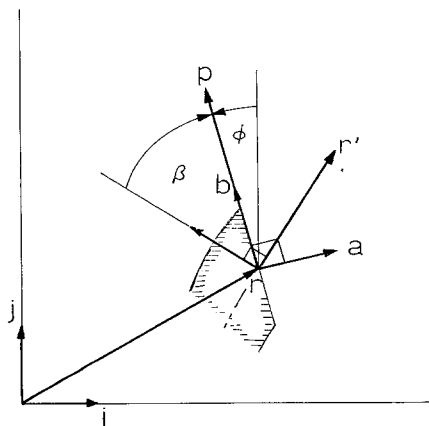


Figure 1.

Let $\mathbf{i}$, $\mathbf{j}$ be an orthonormal basis for the plane of $\mathbf{r}$ and $\mathbf{p}$, and let $\mathbf{k} = \mathbf{i} \times \mathbf{j}$. We define geometrical variables $\mathbf{a}(S)$, $\mathbf{b}(S)$, $\phi(S)$, $\xi(S)$, $\eta(S)$, $x(S)$, $y(S)$, $\rho(S)$ by

$$\mathbf{a}(S) = \cos \phi(S) \mathbf{i} + \sin \phi(S) \mathbf{j}, \quad (2.2a)$$

$$\mathbf{b}(S) = -\sin \phi(S) \mathbf{i} + \cos \phi(S) \mathbf{j}, \quad (2.2b)$$

$$\mathbf{r}'(S) \equiv [1 + \xi(S)] \mathbf{a}(S) + \eta(S) \mathbf{b}(S), \quad (2.3a)$$

$$\mathbf{r}(S) = x(S) \mathbf{i} + y(S) \mathbf{j}, \quad (2.3b)$$

$$\mathbf{p}(S) \equiv [1 + \rho(S)] \mathbf{b}(S). \quad (2.4)$$

Thus $\phi(S)$ gives the angle between the deformed cross section and the fixed $\mathbf{j}$ -axis and $\rho(S)$ gives the extension suffered by the cross section. The stretch along the axis is

$$|\mathbf{r}'(S)| = \{[1 + \xi(S)]^2 + \eta(S)^2\}^{1/2} \equiv 1 + \varepsilon(S) \quad (2.5a)$$

and the shear angle $\beta(S)$ representing the reduction in angle between the axis at S and cross section at S is defined by

$$\sin \beta(S) = \mathbf{b}(S) \cdot \mathbf{r}'(S) / |\mathbf{r}'(S)| = \eta(S) \{[1 + \xi(S)]^2 + \eta(S)^2\}^{-1/2}. \quad (2.5b)$$

The *strain variables* for our problem are the components of

$$\mathbf{w}(S) \equiv (\xi(S), \eta(S), \rho(S), \sigma(S) \equiv \rho'(S), \mu(S) \equiv \phi'(S)). \quad (2.6)$$

The requirement that the reference configuration have a straight axis lying along the $\mathbf{i}$ -axis and have cross sections perpendicular to this axis is ensured by the equalities $\eta = 0$, $\rho' = 0$, $\phi' = 0$ in this configuration. We assume further that $\xi = 0$, $\rho = 0$ in this configuration.

We require that

$$\mathbf{k} \cdot [\mathbf{r}'(S) \times \mathbf{p}(S)] \equiv [1 + \xi(S)][1 + \rho(S)] > 0 \quad (2.7)$$

so that $|\mathbf{r}'(S)| > 0$ and $|\mathbf{p}(S)| > 0$, which imply that no portion of the axis or the cross section can be squashed to zero length, and so that the cross section is never collinear with the axis.

We let $\hat{\Xi}$, $\hat{H}$, $\hat{P}$, $\hat{\Sigma}$, $\hat{M}$ denote the generalized forces corresponding to the generalized displacements ξ , η , ρ , σ , μ (according to a suitable principle of virtual work). $\hat{\Xi}(S)$ represents the component of resultant force at S in the $\mathbf{a}(S)$ direction, $\hat{H}(S)$ represents the component of resultant force at S in the $\mathbf{b}(S)$ direction, and $\hat{M}(S)$ represents the bending moment at S . $\hat{P}(S)$ and $\hat{\Sigma}(S)$ are director forces at S . If the rod is subjected only to loads applied at its ends, then its equilibrium is governed by the equations

$$\hat{\Xi}' - \hat{H}\phi' = 0, \quad (2.8)$$

$$\hat{H}' + \hat{\Xi}\phi' = 0, \quad (2.9)$$

$$\hat{\Sigma}' - \hat{M}(1 + \rho)^{-1}\phi' - \hat{P} = 0, \quad (2.10)$$

$$\hat{M} + \hat{H}(1 + \xi) - \hat{\Xi}\eta = 0. \quad (2.11)$$

Equations (2.8), (2.9), (2.11) are equivalent to the standard expressions for balance of force and moments for static problems. A full discussion of these variables is given in [1]. Also see the Appendix in Section 8.

We introduce constitutive functions

$$(\mathbf{w}, S) \equiv (\xi, \eta, \rho, \sigma, \mu, S) \mapsto \Xi(\mathbf{w}, S), H(\mathbf{w}, S), P(\mathbf{w}, S), \Sigma(\mathbf{w}, S), M(\mathbf{w}, S). \quad (2.12)$$

These functions are defined only for $\xi \in (-1, \infty)$, $\rho \in (-1, \infty)$ in accordance with (2.7). We assume that these functions are four times continuously differentiable. We assume that the material of the rod is homogeneous so that these functions are independent of S . In this case our constitutive equations have the form:

$$\begin{aligned} \hat{\Xi}(S) &= \Xi(\mathbf{w}(S)), & \hat{H}(S) &= H(\mathbf{w}(S)), & \hat{P}(S) &= P(\mathbf{w}(S)), & \hat{\Sigma}(S) &= \Sigma(\mathbf{w}(S)), \\ \hat{M}(S) &= M(\mathbf{w}(S)). \end{aligned} \quad (2.13)$$

These constitutive relations give the most general form of elastic response consistent with frame-indifference and planarity for materials with independent constitutive variables $\mathbf{r}$, $\mathbf{r}'$, $\mathbf{p}$, $\mathbf{p}'$. (cf. [1].)

We require that

$$\begin{pmatrix} \partial \Xi / \partial \xi & \partial H / \partial \xi & \partial \Sigma / \partial \xi & \partial M / \partial \xi \\ \partial \Xi / \partial \eta & \partial H / \partial \eta & \partial \Sigma / \partial \eta & \partial M / \partial \eta \\ \partial \Xi / \partial \sigma & \partial H / \partial \sigma & \partial \Sigma / \partial \sigma & \partial M / \partial \sigma \\ \partial \Xi / \partial \mu & \partial H / \partial \mu & \partial \Sigma / \partial \mu & \partial M / \partial \mu \end{pmatrix} \quad (2.14a)$$

be positive definite and that

$$\partial P / \partial \rho > 0. \quad (2.14b)$$

These conditions imply that more tensile force produces more extension, more bending moment produces more flexure, etc. The genesis of these conditions, which are the one-dimensional analogs of the strong ellipticity condition for the three-dimensional theory, is examined at length in [6, Section 3]. (For an interesting discussion of material instabilities that can occur in the absence of such monotonicity conditions, see [17, 18].) We impose the following growth conditions:

$$\Xi \rightarrow -\infty \quad \text{as} \quad \xi \rightarrow -1, \quad \Xi \rightarrow \infty \quad \text{as} \quad \xi \rightarrow \infty, \quad (2.15a)$$

$$H \rightarrow \pm\infty \quad \text{as} \quad \eta \rightarrow \pm\infty, \quad (2.15b)$$

$$P \rightarrow -\infty \quad \text{as} \quad \rho \rightarrow -1, \quad P \rightarrow \infty \quad \text{as} \quad \rho \rightarrow \infty, \quad (2.15c)$$

$$\Sigma \rightarrow \pm\infty \quad \text{as} \quad \sigma \rightarrow \pm\infty, \quad (2.15d)$$

$$M \rightarrow \pm\infty \quad \text{as} \quad \mu \rightarrow \pm\infty. \quad (2.15e)$$

We also require that the cross section of the rod possess sufficient symmetry for it to be no easier to shear or bend the rod in one sense than in the opposite sense. This vague statement is made precise by the following requirements on the constitutive functions:

$$\Xi(\xi, \eta, \rho, \sigma, \mu) = \Xi(\xi, \eta, \rho, -\sigma, -\mu) = \Xi(\xi, -\eta, \rho, \sigma, -\mu) = \Xi(\xi, -\eta, \rho, -\sigma, \mu), \quad (2.16a)$$

$$H(\xi, \eta, \rho, \sigma, \mu) = H(\xi, \eta, \rho, -\sigma, -\mu) = -H(\xi, -\eta, \rho, \sigma, -\mu) = -H(\xi, -\eta, \rho, -\sigma, \mu), \quad (2.16b)$$

$$P(\xi, \eta, \rho, \sigma, \mu) = P(\xi, \eta, \rho, -\sigma, -\mu) = P(\xi, -\eta, \rho, \sigma, -\mu) = P(\xi, -\eta, \rho, -\sigma, \mu), \quad (2.16c)$$

$$\Sigma(\xi, \eta, \rho, \sigma, \mu) = -\Sigma(\xi, \eta, \rho, -\sigma, -\mu) = \Sigma(\xi, -\eta, \rho, \sigma, -\mu) = -\Sigma(\xi, -\eta, \rho, -\sigma, \mu), \quad (2.16d)$$

$$M(\xi, \eta, \rho, \sigma, \mu) = -M(\xi, \eta, \rho, -\sigma, -\mu) = -M(\xi, -\eta, \rho, \sigma, -\mu) = M(\xi, -\eta, \rho, -\sigma, \mu), \quad (2.16e)$$

The justification of these conditions is obtained in the Appendix, Section 8.

We consider a special system of boundary conditions:

$$\mathbf{r}(0) = \mathbf{0}, \quad \mathbf{r}(1) = (1 + \gamma)\mathbf{i} \quad \text{with} \quad \gamma > 0, \quad (2.17a)$$

$$\Sigma(0) = 0, \quad \Sigma(1) = 0, \quad (2.17b)$$

$$M(0) = 0, \quad M(1) = 0. \quad (2.17c)$$

Condition (2.17a) says that the left end is fixed and the right end is displaced a distance γ in the $\mathbf{i}$ -direction. Conditions (2.17b), (2.17c) say that each end is hinged in such a way that no resistance is offered to changes in thickness. The positive-definiteness of (2.14) in conjunction with (2.15d,e) implies by a standard degree theoretic argument that the algebraic equations $\Sigma(\xi, \eta, \rho, \sigma, \mu) = 0$, $M(\xi, \eta, \rho, \sigma, \mu) = 0$ can be solved uniquely for σ, μ in terms of the other variables and (2.16d,e) implies that this solution is $(\sigma, \mu) = (0, 0)$. Thus (2.17b,c) are equivalent to

$$\rho'(0) = 0, \quad \rho'(1) = 0, \quad (2.17d)$$

$$\phi'(0) = 0, \quad \phi'(1) = 0. \quad (2.17e)$$

We shall also comment briefly on the alternative set of boundary conditions in which (2.17c) (or its equivalent (2.17e)) is replaced by

$$\phi(0) = 0, \quad \phi(1) = 0. \quad (2.18)$$

This set of conditions corresponds to supports that fix the orientation of the end sections but offer no resistance to thickness changes.

Summary. Our boundary value problem consists of the geometric relations (2.2)–(2.5), the equilibrium equations (2.8)–(2.11), the constitutive relations (2.13), and the boundary conditions (2.17).

3. Trivial solutions

The boundary value problem has a solution with $\xi = \xi_0$ (const), $\eta = \eta_0$ (const), $\rho = \rho_0$ (const), $\phi = \phi_0$ (const) provided

$$1 + \xi_0 = (1 + \gamma) \cos \phi_0, \quad \eta_0 = -(1 + \gamma) \sin \phi_0, \quad (3.1)$$

$$H(\xi_0, \eta_0, \rho_0, 0, 0)(1 + \xi_0) - \Xi(\xi_0, \eta_0, \rho_0, 0, 0)\eta_0 = 0, \quad (3.2)$$

$$P(\xi_0, \eta_0, \rho_0, 0, 0) = 0. \quad (3.3)$$

Conditions (2.14b) and (2.15c) imply that (3.3) can be solved uniquely for ρ_0 as a function $\hat{\rho}$ of ξ_0 and η_0 . Then the existence of such a solution devolves upon finding a solution ϕ_0 of the single equation

$$H \cos \phi_0 + \Xi \sin \phi_0 = 0, \quad (3.4a)$$

where the arguments of H and Ξ are

$$(1 + \gamma) \cos \phi_0 - 1, -(1 + \gamma) \sin \phi_0, \hat{\rho}((1 + \gamma) \cos \phi_0 - 1, -(1 + \gamma) \sin \phi_0), 0, 0. \quad (3.4b)$$

The expression in (3.4a) is just the component of the resultant force in the direction normal to the axis. Relations (2.14) and (2.15) are of little help in determining the number of solutions of (3.4). But (2.16a,b) imply that if ϕ_0 is a solution, then so is $-\phi_0$. From this and (2.16c) we can conclude that if $(\xi_0, \eta_0, \rho_0, \phi_0)$ satisfies (3.1)–(3.3) then so does $(\xi_0, -\eta_0, \rho_0, -\phi_0)$.

These remarks ensure however that (3.1)–(3.3) has at least one solution

$$\xi_0 = \gamma, \quad \eta_0 = 0, \quad \rho_0 = \hat{\rho}(\gamma, 0) \equiv \bar{\rho}(\gamma), \quad \phi_0 = 0 \quad (3.5)$$

corresponding to a straight elongated axis and unsheared cross sections with uniform thickness. In Sections 5 and 6 we shall examine bifurcation from this trivial solution.

We finally note that if (2.17c) is replaced by (2.18), then the only solution with constant strain is (3.5).

4. The perturbation scheme

We treat our bifurcation problem by a convergent perturbation scheme that relates the solution to the total elongation γ . Since we cannot expect the solution to be a single-valued function of γ , we introduce a new parameter α and assume that both γ and the geometric variables have expansions of the form

$$\gamma(\alpha) = \gamma_0 + \gamma_1 \alpha + \gamma_2 \alpha^2/2! + \gamma_3 \alpha^3/3! + O(\alpha^4), \quad (4.1)$$

$$\xi(S; \alpha) = \xi_0 + \xi_1(S) \alpha + \xi_2(S) \alpha^2/2! + \xi_3(S) \alpha^3/3! + O(\alpha^4), \quad \text{etc.} \quad (4.2)$$

The validity of these representations is discussed in Section 5.

We substitute (4.1), (4.2) into the equations of our boundary value problem. We obtain equations for $\{\gamma_k, \xi_k, \dots, k = 1, 2, 3\}$ by differentiating these equations repeatedly with respect to α and then setting $\alpha = 0$. If we set

$$\Xi_0 = \Xi(\mathbf{w}_0) \equiv \Xi(\xi_0, \eta_0, \rho_0, 0, 0), \quad H_0 = H(\mathbf{w}_0) \equiv H(\xi_0, \eta_0, \rho_0, 0, 0), \quad (4.3)$$

$$\Xi_{w_i} w_k^i = \sum_{i=1}^5 \frac{\partial \Xi}{\partial w^i}(\mathbf{w}_0) w_k^i, \quad \text{etc., with } (w_k^1, w_k^2, w_k^3, w_k^4, w_k^5) = (\xi_k, \eta_k, \rho_k, \sigma_k, \mu_k), \quad (4.4)$$

$$\Xi_\xi = \partial \Xi(\mathbf{w}_0) / \partial \xi, \quad \text{etc.,} \quad (4.4b)$$

then for $k = 1, 2, 3$ we obtain

$$(\Xi_{w^i} w_k^i)' - H_0 \phi_k' = F_{1k}, \quad (4.5)$$

$$(H_{w^i} w_k^i)' + \Xi_0 \phi_k' = F_{2k}, \quad (4.6)$$

$$(\Sigma_{w^i} w_k^i)' - P_{w^i} w_k^i = F_{3k}, \quad (4.7)$$

$$(M_{w^i} w_k^i)' + H_{w^i} w_k^i (1 + \xi_0) + H_0 \xi_k - \Xi_{w^i} w_k^i \eta_0 - \Xi_0 \eta_k = F_{4k}, \quad (4.8)$$

$$\int_0^1 [\xi_k \cos \phi_0 - (1 + \xi_0) \sin \phi_0 \phi_k - \eta_k \sin \phi_0 - \eta_0 \cos \phi_0 \phi_k] dS = \gamma_k + F_{5k}, \quad (4.9)$$

$$\int_0^1 [\xi_k \sin \phi_0 + (1 + \xi_0) \cos \phi_0 \phi_k + \eta_k \cos \phi_0 - \eta_0 \sin \phi_0 \phi_k] dS = F_{6k}, \quad (4.10)$$

$$\rho_k'(0) = 0, \quad \rho_k'(1) = 0, \quad (4.11)$$

$$\phi_k'(0) = 0, \quad \phi_k'(1) = 0, \quad (4.12)$$

where $\{F_{lk}(S)\}$ have the form

$$F_{lk}(S) = \hat{F}_{lk}(\{w_j(S), \gamma_j\}), \quad \text{for } l = 1, 2, 3, 4, k = 1, 2, 3, j < k, \quad (4.13)$$

$$F_{lk} = \hat{F}_{lk}(\{w_j, \gamma_j\}), \quad \text{for } l = 5, 6, k = 1, 2, 3, j < k. \quad (4.14)$$

Also, $F_{11} = 0$ for $l = 1, \dots, 6$. In (4.3)–(4.6), we have not yet used the full force of the symmetry relations (2.16). Note that (4.9) and (4.10) come from (2.17a).

5. Bifurcation from the straight unsheared state

Let $\mathbf{w}_0$ be the trivial solution given by (3.5). By using (2.16) we can reduce (4.5)–(4.12) for $k = 1$ to an uncoupled pair of boundary value problems for third order systems:

$$[H_\eta \eta_1 + \Xi_0 \phi_1]' = 0, \quad (5.1a)$$

$$M_\mu \phi_1'' + [(1 + \gamma_0) H_\eta - \Xi_0] \eta_1 = 0, \quad (5.1b)$$

$$\phi_1'(0) = 0, \quad \phi_1'(1) = 0, \quad (5.1c)$$

$$\int_0^1 [(1 + \gamma_0) \phi_1 + \eta_1] dS = 0, \quad (5.1d)$$

$$[\Xi_\xi \xi_1 + \Xi_\rho \rho_1]' = 0, \quad (5.2a)$$

$$\Sigma_\sigma \rho_1'' - P_\xi \xi_1 - P_\rho \rho_1 = 0, \quad (5.2b)$$

$$\rho_1'(0) = 0, \quad \rho_1'(1) = 0, \quad (5.2c)$$

$$\int_0^1 \xi_1 dS = \gamma_1. \quad (5.2d)$$

Note that (5.2) is not a homogeneous problem. (This is a consequence of our formulation in which the eigenvalue parameter γ appears as a nonhomogeneous term.)

We start with (5.1). We can readily show that this system has a nontrivial solution if and only if there is an integer m such that

$$p(\gamma_0) \equiv -\Xi_0[(1 + \gamma_0)H_\eta - \Xi_0]/H_\eta M_\mu = m^2 \pi^2, \quad \Xi_0 \neq 0. \quad (5.3)$$

Roots of this equation other than $\gamma_0 = 0$ are called *eigenvalues of shear type*. (For certain materials there may not be any.) If (5.3) holds, then the solution of (5.1) has the form

$$(\eta_1(S), \phi_1(S)) = \text{const.} (1, -H_\eta/\Xi_0) \cos m\pi S. \quad (5.4)$$

When $\eta_1 \neq 0$, we can fix this constant and simplify later computations by defining

$$\alpha = \int_0^1 \eta(S, \alpha) \eta_1(S) dS / \sqrt{\int_0^1 [\eta_1(S)]^2 dS}, \quad (5.5)$$

which implies that

$$1 = \int_0^1 [\eta_1(S)]^2 dS \quad (5.6)$$

and

$$0 = \int_0^1 \eta_k(S) \eta_1(S) dS \quad \text{for } k > 1. \quad (5.7)$$

(cf. [11].) Equation (5.6) makes the constant of (5.4) equal to unity; thus (5.4) becomes

$$\eta_1(S) = \cos m\pi S, \quad \phi_1(S) = -(H_\eta/\Xi_0) \cos m\pi S. \quad (5.8)$$

Let us examine (5.3). We have $p(0) = 0$. The definition of $\bar{p}$ implies that

$$\frac{dp}{d\gamma_0} = -[(\Xi_\xi P_\rho - \Xi_\rho P_\xi)/P_\rho M_\mu] \big|_{\gamma_0=0}. \quad (5.9)$$

The right side of (5.9) is determined from the linear theory of elastic rods. In most of these theories (in particular, in all the rod theories corresponding to three-dimensional theories with monotone stress-strain laws, which include all problems

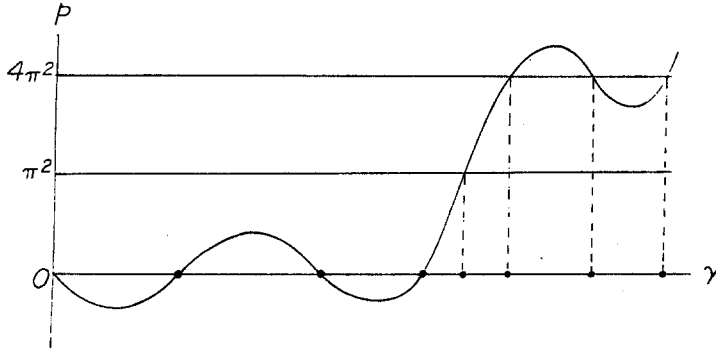


Figure 2. Eigenvalues of shear type.

with a convex strain energy function), the right side is negative. In this case p has the form illustrated in Figure 2.

We conclude that (i) the smallest eigenvalue corresponds to $m = 0$, (ii) the smallest root of $p(\gamma_0) = m^2 \pi^2$ is smaller than the smallest root of $p(\gamma_0) = (m+1)^2 \pi^2$, and (iii) the only eigenvalues that can be nearest to a root of $p(\gamma_0) = m^2 \pi^2$ are roots of $p(\gamma_0) = (m-1)^2 \pi^2$, $p(\gamma_0) = m^2 \pi^2$, $p(\gamma_0) = (m+1)^2 \pi^2$.

We now turn to (5.2). We find that (5.2a,b) implies that

$$\rho_1'' + q(\gamma_0)\rho = P_\xi C / \Sigma_\sigma, \quad (5.10)$$

where

$$q(\gamma_0) = (P_\xi \Xi_\rho - P_\rho \Xi_\xi) / \Xi_\xi \Sigma_\sigma \quad (5.11)$$

and where C is an integration constant for (5.2a). We find the following results. (i) If $q(\gamma_0) \neq \bar{n}^2 \pi^2$, where $\bar{n}$ is any nonzero integer, then (5.2) has the solution

$$\xi_1 = \gamma_1, \quad \rho_1 = P_\xi \gamma_1 / P_\rho. \quad (5.12)$$

(ii) If $q(\gamma_0) = \bar{n}^2 \pi^2$, where $\bar{n}$ is a nonzero integer, then (5.2) has solutions of the form

$$\begin{aligned} \xi_1(S) &= -a(\Xi_\rho / \Xi_\xi) \cos \bar{n} \pi S + \gamma_1, \\ \rho_1(S) &= a \cos \bar{n} \pi S - P_\xi \gamma_1 / P_\rho, \end{aligned} \quad (5.13)$$

where a is arbitrary. $\bar{n} \pi a$ effectively represents the ratio of the amplitude of ρ_1' to the amplitude of η . Indeed, when $\rho_1' \neq 0$, we may set

$$(m\pi)a\alpha = \int_0^1 [\rho'(S, \alpha)\rho_1'(S)] dS / \sqrt{\int_0^1 [\rho_1'(S)]^2 dS}, \quad (5.14)$$

which implies that

$$(m\pi)^2 a^2 = \int_0^1 [\rho_1'(S)]^2 dS \quad (5.15)$$

and

$$0 = \int_0^1 \rho'_k(S) \rho'_1(S) dS \quad \text{for } k > 1. \quad (5.16)$$

Values of γ_0 satisfying $q(\gamma_0) = \bar{n}^2 \pi^2$ are called *eigenvalues of necking type*.

The value of $q(0)$ is determined from the linear theory. If this has monotone stress-strain laws, then $q(0) < 0$. Except for this fact, we can expect q to have the same character as p , which is indicated in Figure 2.

We now turn to the second perturbation. As before, we can use (2.16) to reduce (4.5)–(4.12) for $k = 2$ to an uncoupled pair of boundary value problems:

$$[H_\eta \eta_2 + \Xi_0 \phi_2]' = -2[H_{\xi\eta} \xi_1 \eta_1 + H_{\rho\eta} \rho_1 \eta_1 + H_{\sigma\mu} \rho'_1 \phi'_1]' - 2[\Xi_\xi \xi_1 + \Xi_\rho \rho_1] \phi'_1, \quad (5.17a)$$

$$\begin{aligned} M_\mu \phi''_2 + [(1 + \gamma_0) H_\eta - \Xi_0] \eta_2 = & -2[M_{\xi\mu} \xi_1 \phi'_1 + M_{\eta\sigma} \eta_1 \rho'_1 + M_{\rho\mu} \rho_1 \phi'_1]' \\ & -2(1 + \gamma_0)[H_{\xi\eta} \xi_1 \eta_1 + H_{\eta\rho} \eta_1 \rho_1 + H_{\sigma\mu} \rho'_1 \phi'_1] \\ & -2H_\eta \eta_1 \xi_1 + 2[\Xi_\xi \xi_1 + \Xi_\rho \rho_1] \eta_1, \end{aligned} \quad (5.17b)$$

$$\phi'_2(0) = 0, \quad \phi'_2(1) = 0, \quad (5.17c)$$

$$\int_0^1 [(1 + \gamma_0) \phi_2 + \eta_2] dS = -2 \int_0^1 \xi_1 \phi_1 dS, \quad (5.17d)$$

$$\int_0^1 \eta_2 \eta_1 dS = 0; \quad (5.17e)$$

$$\begin{aligned} [\Xi_\xi \xi_2 + \Xi_\rho \rho_2]' = & -[\Xi_{\xi\xi} \xi_1^2 + 2\Xi_{\xi\rho} \xi_1 \rho_1 + \Xi_{\eta\eta} \eta_1^2 + \Xi_{\rho\rho} \rho_1^2 \\ & + \Xi_{\sigma\sigma} (\rho'_1)^2 + \Xi_{\mu\mu} (\phi'_1)^2]' + 2H_\eta \eta_1 \phi'_1, \end{aligned} \quad (5.18a)$$

$$\begin{aligned} \Sigma_{\sigma\sigma} \rho''_2 - P_\xi \xi_2 - P_\rho \rho_2 = & -2[\Sigma_{\xi\sigma} \xi_1 \rho'_1 + \Sigma_{\eta\mu} \eta_1 \phi'_1 + \Sigma_{\rho\sigma} \rho_1 \rho'_1]' + 2M_\mu (\phi'_1)^2 / (1 + \rho_0) \\ & + P_{\xi\xi} \xi_1^2 + 2P_{\xi\rho} \xi_1 \rho_1 + P_{\eta\eta} \eta_1^2 + P_{\rho\rho} \rho_1^2 + P_{\sigma\sigma} (\rho'_1)^2 + P_{\mu\mu} (\phi'_1)^2, \end{aligned} \quad (5.18b)$$

$$\rho'_2(0) = 0, \quad \rho'_2(1) = 0, \quad (5.18c)$$

$$\int_0^1 \xi_2 dS = \gamma_2 + \int_0^1 [(1 + \gamma_0) \phi_1^2 + 2\eta_1 \phi_1] dS, \quad (5.18d)$$

$$\int_0^1 \rho'_2 \rho'_1 dS = 0. \quad (5.18e)$$

We have several distinct cases depending on the values of $p(\gamma_0)$ and $q(\gamma_0)$. These are itemized in the following table in which $\bar{m}$ and $\bar{n}$ represent nonzero integers.

$\begin{array}{c} q \\ \backslash \\ p \end{array}$	$\begin{array}{c} \neq (n\pi)^2 \\ \neq 0 \end{array}$	$= 0$	$= (\bar{n}\pi)^2$
$\begin{array}{c} \neq (m\pi)^2 \\ \neq 0 \end{array}$	Case (X, Y)	Case (X, 0)	Case (X, $\bar{n}$)
$= 0$	Case (0, Y)	Case (0, 0)	Case (0, $\bar{n}$)
$= (\bar{m}\pi)^2$	Case ($\bar{m}$, Y)	Case ($\bar{m}$, 0)	Case ($\bar{m}$, $\bar{n}$)

(5.19)

The case ($\bar{m}$, $\bar{n}$) has a number of important subcases. The cases of especial interest to us are (X, $\bar{n}$), (0, Y), and ($\bar{m}$, Y).

The principal tool for the efficient analysis of these cases is the following special version of the alternative theorem: If the boundary value problem

$$u'' + k^2 \pi^2 u = f, \quad u'(0) = 0, \quad u'(1) = 0$$

has a solution for integral k , then

$$\int_0^1 f(S) \cos k\pi S \, dS = 0.$$

We obtain problems of this form by integrating (5.17a) and (5.18a) and then substituting the resulting expressions into (5.17b) and (5.18b), respectively.

If $p(\gamma_0) \neq (m\pi)^2$, then $\eta_1 = 0$, $\phi_1 = 0$ and the nonhomogeneous terms of (5.17) vanish. It follows that $\eta_2 = 0$, $\phi_2 = 0$. Thus cases (X, Y), (X, 0), (X, $\bar{n}$) have no shearing. In cases (X, Y) and (X, 0) by substituting (5.12) into (5.18) we find that either

$$\gamma_1 = 0 \quad \text{so that} \quad \xi_1 = 0, \quad \rho_1 = 0, \quad \xi_2 = \gamma_2, \quad \rho_2 = -P_\xi \gamma_2 / P_\rho \quad (5.20)$$

or

$$\xi_2 = \gamma_2 = -\gamma_1^2 [P_{\xi\xi} - 2P_{\xi\rho}(P_\xi/P_\rho) + P_{\rho\rho}(P_\xi/P_\rho)^2] / P_\xi, \quad \rho_2 = 0. \quad (5.21)$$

The possibility (5.20) leads to a trivial solution for the first perturbation and causes the second perturbation to look like (5.12). Repeated perturbations would produce possibilities of the form (5.20) and (5.21). Possibility (5.21) corresponds either to a trivial solution if $\gamma_1 = 0$ or corresponds to a bifurcation from the original trivial solution to another trivial solution. Thus cases (X, Y) and (X, 0) are of at most minor interest.

In case (X, $\bar{n}$) we substitute (5.13) into (5.18) and use the alternative theorem to conclude that (5.18) has a solution only if

$$\begin{aligned} a\gamma_1 [(\Xi_\xi P_{\xi\xi} - P_\xi \Xi_{\xi\xi}) \Xi_\rho P_\rho - (\Xi_\xi P_{\xi\rho} - P_\xi \Xi_{\xi\rho})(\Xi_\xi P_\rho + \Xi_\rho P_\xi) \\ + (\Xi_\xi P_{\rho\rho} - P_\xi \Xi_{\rho\rho}) \Xi_\xi P_\xi] = 0. \end{aligned} \quad (5.22)$$

We shall assume that the coefficient of $a\gamma_1$ is not zero, which incidentally requires the constitutive functions (Ξ, P) to be nonlinear. (This is a *generic* case, i.e., there is probability zero that a set of admissible constitutive functions chosen at random will have this coefficient = 0. Nevertheless the case in which this coefficient vanishes is not devoid of mathematical and physical interest. We pass over these questions because their treatment would obscure our main objectives.) In this case we obtain $\gamma_1 = 0$, so that we have

$$\xi_1 = -a(\Xi_\rho/\Xi_\xi) \cos \bar{n}\pi S, \quad \rho_1 = a \cos \bar{n}\pi S, \quad \gamma_1 = 0, \quad (5.23a)$$

$$\xi_2 = \gamma_2 + \frac{a^2[A_2 - \bar{n}^2\pi^2\Xi_{\sigma\sigma}]}{2\Xi_\xi} \left[\frac{P_\xi}{3\bar{n}^2\pi^2\Sigma_\sigma\Xi_\xi} - 1 \right] \cos 2\bar{n}\pi S, \quad (5.23b)$$

$$\rho_2 = -P_\xi\gamma_2/P_\rho - a^2[B_2 + \bar{n}^2\pi^2P_{\sigma\sigma}]/2P_\rho - \frac{a^2[B_2 - \bar{n}^2\pi^2P_{\sigma\sigma} - (A_2 - \bar{n}^2\pi^2\Xi_{\sigma\sigma})P_\xi/\Xi_\xi]}{6\bar{n}^2\pi^2\Sigma_\sigma} \cos 2\bar{n}\pi S, \quad (5.23c)$$

where

$$A_2 = \Xi_{\xi\xi}(\Xi_\rho/\Xi_\xi)^2 - 2\Xi_{\xi\rho}(\Xi_\rho/\Xi_\xi) + \Xi_{\rho\rho}, \quad (5.24a)$$

$$B_2 = P_{\xi\xi}(\Xi_\rho/\Xi_\xi)^2 - 2P_{\xi\rho}(\Xi_\rho/\Xi_\xi) + P_{\rho\rho}. \quad (5.24b)$$

In case (0, Y), direct integration shows that if (5.17) has a solution, then

$$\gamma_1[H_\eta + (1 + \gamma_0)(H_{\xi\eta}P_\rho - H_{\eta\rho}P_\xi)/P_\rho + (\Xi_\rho P_\xi - \Xi_\xi P_\rho)/P_\rho] = 0. \quad (5.25)$$

As in (5.22), we assume that the coefficient of γ_1 does not vanish. Then $\gamma_1 = 0$ and (5.8), (5.12), (5.17), (5.18) imply that

$$\xi_1 = 0, \quad \rho_1 = 0, \quad \eta_1 = 1, \quad \phi_1 = -H_\eta/\Xi_0, \quad \gamma_1 = 0, \quad (5.26a)$$

$$\eta_2 = 0, \quad \phi_2 = 0, \quad (5.26b,c)$$

$$\xi_2 = \gamma_2 - H_\eta/\Xi_0, \quad \rho_2 = -(P_\xi/P_\rho)(\gamma_2 - H_\eta/\Xi_0) - P_{\eta\eta}/P_\rho. \quad (5.26d,e)$$

In case $(\bar{m}, Y)$, we use the alternative theorem to show that (5.17) has a solution only if

$$\gamma_1\{H_\eta + \Xi_0(H_{\xi\eta}P_\rho - H_{\rho\eta}P_\xi)/H_\eta P_\rho + [2 - H_\eta(1 + \gamma_0)/\Xi_0](\Xi_\rho P_\xi - \Xi_\xi P_\rho)/P_\rho\} = 0. \quad (5.27)$$

(Note that (5.27) reduces to (5.25) when $\bar{m}$ is replaced by 0.) We assume that the coefficient of γ_1 does not vanish so that $\gamma_1 = 0$ and (5.8), (5.12), (5.17), (5.18) imply that

$$\xi_1 = 0, \quad \rho_1 = 0, \quad \eta_1(S) = \cos \bar{m}\pi S, \quad (5.28a)$$

$$\phi_1(S) = -(H_\eta/\Xi_0) \cos \bar{m}\pi S, \quad \gamma_1 = 0, \quad (5.28a)$$

$$\eta_2 = 0, \quad \phi_2 = 0, \quad (5.28b,c)$$

$$\xi_2 = \gamma_2 + C_2 + \frac{[(P_\rho - 4\bar{m}^2 \pi^2 \Sigma_\sigma)D_2 - \Xi_\rho F_2]}{\Xi_\xi \Sigma_\sigma (q - 4\bar{m}^2 \pi^2)} \cdot \cos 2\bar{m}\pi S, \quad (5.28d)$$

$$\rho_2 = -(P_\xi/P_\rho)(\gamma_2 + C_2) - E_2/P_\rho + \frac{(\Xi_\xi F_2 - P_\xi D_2)}{\Xi_\xi \Sigma_\sigma (q - 4\bar{m}^2 \pi^2)} \cos 2\bar{m}\pi S, \quad (5.28e)$$

where

$$2C_2 = (H_\eta/\Xi_0)[1 + \gamma_0](H_\eta/\Xi_0) - 2], \quad (5.29a)$$

$$2D_2 = \Xi_{\eta\eta} + H_\eta^2/\Xi_0 - (\bar{m}\pi)^2 \Xi_{\mu\mu} (H_\eta/\Xi_0)^2, \quad (5.29b)$$

$$2E_2 = P_{\eta\eta} + (\bar{m}\pi H_\eta/\Xi_0)^2 [P_{\mu\mu} + 2M_\mu/(1 + \rho_0)], \quad (5.29c)$$

$$2F_2 = P_{\eta\eta} - (\bar{m}\pi H_\eta/\Xi_0)^2 [P_{\mu\mu} + 2M_\mu/(1 + \rho_0)] - 4(\bar{m}\pi)^2 (H_\eta/\Xi_0) \Sigma_{\eta\mu}. \quad (5.29d)$$

In case (0, 0) direct integration yields (5.25) and (5.26). In case $(\bar{m}, 0)$, the alternative theorem and direct integration yields the obvious specializations of (5.27) and (5.28). The cases $(0, \bar{n})$ and $(\bar{m}, \bar{n})$ are handled the same, but various subcases require special treatment. We do not examine these cases since they are not generic. (Neither are the cases (0, 0) and $(\bar{m}, 0)$.)

We now turn to the third perturbation. We again use (2.16) to reduce (4.5)–(4.12) for $k = 3$ to an uncoupled pair of boundary value problems:

$$\begin{aligned} [H_\eta \eta_3 + \Xi_0 \phi_3]' = & -2[H_{\xi\eta}(\xi_1 \eta_2 + \xi_2 \eta_1) + H_{\rho\eta}(\rho_1 \eta_2 + \rho_2 \eta_1) + H_{\sigma\mu}(\rho'_1 \phi'_2 + \rho'_2 \phi'_1)]' \\ & - [3H_{\xi\xi\eta} \xi_1^2 \eta_1 + 6H_{\xi\eta\rho} \xi_1 \eta_1 \rho_1 + 6H_{\xi\sigma\mu} \xi_1 \rho'_1 \phi'_1 \\ & + H_{\eta\eta\eta} (\eta_1)^3 + 3H_{\eta\rho\rho} \eta_1 \rho_1^2 \\ & + 3H_{\eta\sigma\sigma} \eta_1 \rho'_1 \rho'_1 + 3H_{\eta\mu\mu} \eta_1 \phi'_1 \phi'_1 + 6H_{\rho\sigma\mu} \rho_1 \rho'_1 \phi'_1]' \\ & - 3[\Xi_\xi \xi_2 + \Xi_\rho \rho_2 + \Xi_{\xi\xi} \xi_1^2 + 2\Xi_{\xi\rho} \xi_1^2 \rho_1 + \Xi_{\eta\eta} \eta_1^2 + \Xi_{\rho\rho} \rho_1^2 \\ & + \Xi_{\sigma\sigma} (\rho'_1)^2 + \Xi_{\mu\mu} (\phi'_1)^2] \phi'_1 - 3[\Xi_{\xi\xi} \xi_1 + \Xi_{\rho\rho} \rho_1] \phi'_2. \end{aligned} \quad (5.30a)$$

$$\begin{aligned} M_\mu \phi_3'' + [(1 + \gamma_0)H_\eta - \Xi_0] \eta_3 \\ = & -3[M_{\xi\mu}(\xi_1 \phi'_2 + \xi_2 \phi'_1) + M_{\eta\sigma}(\eta_1 \rho'_2 + \eta_2 \rho'_1) + M_{\rho\mu}(\rho_1 \phi'_2 + \rho_2 \phi'_1)]' \\ & - [3M_{\eta\eta\mu} \eta_1^2 \phi'_1 + 3M_{\xi\xi\mu} \xi_1^2 \phi'_1 + 6M_{\xi\eta\sigma} \xi_1 \eta_1 \rho'_1 + 6M_{\xi\rho\mu} \xi_1 \rho_1 \phi'_1 + 6M_{\eta\rho\sigma} \eta_1 \rho_1 \rho'_1 \\ & + 3M_{\rho\rho\mu} \rho_1^2 \phi'_1 + 3M_{\sigma\sigma\mu} (\rho'_1)^2 \phi'_1 + M_{\mu\mu\mu} (\phi'_1)^3]' \\ & - 3(1 + \gamma_0)[H_{\xi\eta}(\xi_1 \eta_2 + \xi_2 \eta_1) + H_{\rho\eta}(\rho_1 \eta_2 + \rho_2 \eta_1) + H_{\sigma\mu}(\rho'_1 \phi'_2 + \rho'_2 \phi'_1)] \\ & - (1 + \gamma_0)[3H_{\xi\xi\eta} \xi_1^2 \eta_1 + 6H_{\xi\eta\rho} \xi_1 \eta_1 \rho_1 + 6H_{\xi\sigma\mu} \xi_1 \rho'_1 \phi'_1 + H_{\eta\eta\eta} \eta_1^3 + 3H_{\eta\rho\rho} \eta_1 \rho_1^2 \\ & + 3H_{\eta\sigma\sigma} \eta_1 (\rho'_1)^2 + 3H_{\eta\mu\mu} \eta_1 (\phi'_1)^2 + 6H_{\rho\sigma\mu} \rho_1 \rho'_1 \phi'_1] \\ & - 3[H_\eta \eta_2 + 2H_{\xi\eta} \xi_1 \eta_1 + 2H_{\rho\eta} \rho_1 \eta_1 + 2H_{\sigma\mu} \rho'_1 \phi'_1] \xi_1 - 3H_\eta \eta_1 \xi_2 \\ & + 3[\Xi_{\xi\xi} \xi_2 + \Xi_{\rho\rho} \rho_2 + \Xi_{\xi\xi} \xi_1^2 + 2\Xi_{\xi\rho} \xi_1^2 \rho_1 + \Xi_{\eta\eta} \eta_1^2 + \Xi_{\rho\rho} \rho_1^2 + \Xi_{\sigma\sigma} (\rho'_1)^2 + \Xi_{\mu\mu} (\phi'_1)^2] \eta_1 \\ & + 3[\Xi_{\xi\xi} \xi_1 + \Xi_{\rho\rho} \rho_1] \eta_2, \end{aligned} \quad (5.30b)$$

$$\phi'_3(0) = 0, \quad \phi'_3(1) = 0, \quad (5.30c)$$

$$\int_0^1 [(1 + \gamma_0)\phi_3 + \eta_3] dS = \int_0^1 [(1 + \gamma_0)\phi_1^3 + 3\eta_1\phi_1^2 - 3\xi_2\phi_1 - 3\xi_1\phi_2] dS, \quad (5.30d)$$

$$\int_0^1 \eta_3\eta_1 dS = 0; \quad (5.30e)$$

$$\begin{aligned} & [\Xi_\xi \xi_3 + \Xi_\rho \rho_3]' \\ &= -3[\Xi_{\xi\xi}\xi_1\xi_2 + \Xi_{\xi\rho}(\xi_1\rho_2 + \xi_2\rho_1) + \Xi_{\eta\eta}\eta_1\eta_2 + \Xi_{\rho\rho}\rho_1\rho_2 + \Xi_{\sigma\sigma}\rho_1'\rho_2' + \Xi_{\mu\mu}\phi_1'\phi_2'] \\ & \quad - [\Xi_{\xi\xi\xi}(\xi_1)^3 + 3\Xi_{\xi\xi\rho}\xi_1^2\rho_1 + 3\Xi_{\xi\eta\eta}\xi_1\eta_1^2 + 3\Xi_{\xi\rho\rho}\xi_1\rho_1^2 + 3\Xi_{\xi\sigma\sigma}\xi_1(\rho_1')^2 \\ & \quad + 3\Xi_{\xi\mu\mu}\xi_1(\phi_1')^2 + 3\Xi_{\rho\eta\eta}\rho_1\eta_1^2 + \Xi_{\rho\rho\rho}(\rho_1)^3 + 3\Xi_{\rho\sigma\sigma}\rho_1(\rho_1')^2 + 3\Xi_{\rho\mu\mu}\rho_1(\phi_1')^2] \\ & \quad + 3H_\eta[\eta_2\phi_1' + \eta_1\phi_2'] + 6[H_{\xi\eta}\xi_1\eta_1 + H_{\rho\eta}\rho_1\eta_1 + H_{\sigma\mu}\rho_1'\phi_1']\phi_1', \end{aligned} \quad (5.31a)$$

$$\begin{aligned} & \Sigma_\sigma \rho_3'' - P_\xi \xi_3 - P_\rho \rho_3 \\ &= -3[\Sigma_{\xi\sigma}(\xi_1\rho_2' + \xi_2\rho_1') + \Sigma_{\eta\mu}(\eta_1\phi_2' + \eta_2\phi_1') + \Sigma_{\rho\sigma}(\rho_1\rho_2' + \rho_2\rho_1')] \\ & \quad - [3\Sigma_{\xi\xi\sigma}\xi_1^2\rho_1' + 6\Sigma_{\xi\eta\mu}\xi_1\eta_1\phi_1' + 6\Sigma_{\xi\rho\sigma}\xi_1\rho_1\rho_1' + 3\Sigma_{\eta\eta\sigma}\eta_1^2\rho_1' + 6\Sigma_{\eta\rho\mu}\eta_1\rho_1\phi_1' \\ & \quad + 3\Sigma_{\rho\rho\sigma}\rho_1^2\rho_1' + \Sigma_{\sigma\sigma\sigma}(\rho_1')^3 + 3\Sigma_{\sigma\mu\mu}\rho_1'(\phi_1')^2] + 6(1 + \rho_0)^{-1}M_\mu\phi_1'\phi_2' \\ & \quad + 6(1 + \rho_0)^{-1}[M_{\xi\mu}\xi_1\phi_1' + M_{\eta\sigma}\eta_1\rho_1' + M_{\rho\mu}\rho_1\phi_1']\phi_1' - 6(1 + \rho_0)^{-2}M_\mu(\phi_1')^2\rho_1 \\ & \quad + 3[P_{\xi\xi}\xi_1\xi_2 + P_{\xi\rho}(\xi_1\rho_2 + \xi_2\rho_1) + P_{\eta\eta}\eta_1\eta_2 + P_{\rho\rho}\rho_1\rho_2 + P_{\sigma\sigma}\rho_1'\rho_2' + P_{\mu\mu}\phi_1'\phi_2'] \\ & \quad + [P_{\xi\xi\xi}(\xi_1)^3 + 3P_{\xi\xi\rho}(\xi_1)^2\rho_1 + 3P_{\xi\eta\eta}\xi_1(\eta_1)^2 + 3P_{\xi\rho\rho}\xi_1(\rho_1)^2 + 3P_{\xi\sigma\sigma}\xi_1(\rho_1')^2 \\ & \quad + 3P_{\xi\mu\mu}\xi_1(\phi_1')^2 + 3P_{\rho\eta\eta}\rho_1(\eta_1)^2 + P_{\rho\rho\rho}(\rho_1)^3 + 3P_{\rho\sigma\sigma}\rho_1(\rho_1')^2 + 3P_{\rho\mu\mu}\rho_1(\phi_1')^2], \end{aligned} \quad (5.31b)$$

$$\rho_3'(0) = 0, \quad \rho_3'(1) = 0, \quad (5.31c)$$

$$\int_0^1 \xi_3 dS = \gamma_3 + \int_0^1 [3\xi_1\phi_1^2 + 3(1 + \gamma_0)\phi_1\phi_2 + 3(\eta_2\phi_1 + \eta_1\phi_2)] dS, \quad (5.31d)$$

$$\int_0^1 \rho_3'\rho_1' dS = 0. \quad (5.31e)$$

Fortunately we can get useful information from these equations without undertaking the tedious process of obtaining the full solution. E.g., in the case $(\bar{m}, Y)$ the Alternative Theorem implies that (5.30a,b,c) has a solution only if

$$\begin{aligned} & 3 \int_0^1 (M_{\xi\mu}\xi_2\phi_1' + M_{\eta\sigma}\eta_1\rho_2' + M_{\rho\mu}\rho_2\phi_1')'\eta_1 dS + \int_0^1 [3M_{\eta\eta\mu}\eta_1^2\phi_1' + M_{\mu\mu\mu}(\phi_1')^3]'\eta_1 dS \\ & \quad + 3H_\eta \int_0^1 \eta_1^2\xi_2 dS + 3(\Xi_0/H_\eta) \int_0^1 [H_{\xi\eta}\xi_2\eta_1 + H_{\rho\eta}\rho_2\eta_1 + H_{\sigma\mu}\rho_2'\phi_1']\eta_1 dS \\ & \quad + (\Xi_0/H_\eta) \int_0^1 [H_{\eta\eta\eta}\eta_1^3 + 3H_{\eta\mu\mu}\eta_1(\phi_1')^2]\eta_1 dS \\ & \quad + 3[1 - (1 + \gamma_0)H_\eta/\Xi_0] \int_0^1 \cos 2m\pi S [\Xi_\xi\xi_2 + \Xi_\rho\rho_2 + \Xi_{\eta\eta}\eta_1^2 + \Xi_{\mu\mu}(\phi_1')^2] dS = 0, \end{aligned} \quad (5.32)$$

where η_1 , ϕ_1 , ξ_2 , ρ_2 are given by (5.28). This is an equation for γ_2 in terms of the constitutive functions and their first three derivatives evaluated at the trivial solution.

For the purposes of interpretation, it is useful to compute the $\mathbf{i}$ -component of the resultant force, which is constant and may therefore be evaluated anywhere. We denote this component by Λ :

$$\Lambda = [\Xi(\mathbf{w})\mathbf{a} + H(\mathbf{w})\mathbf{b}] \cdot \mathbf{i} = \Xi(\mathbf{w}) \cos \phi - H(\mathbf{w}) \sin \phi, \quad (5.33)$$

from which we get

$$\Lambda = \Lambda_0 + \alpha \Lambda_1 + (\alpha^2/2!) \Lambda_2 + \dots \quad (5.34a)$$

with

$$\Lambda_0 = \Xi_0, \quad (5.34b)$$

$$\Lambda_1 = \Xi_\xi \xi_1 + \Xi_\rho \rho_1, \quad (5.34c)$$

$$\begin{aligned} \Lambda_2 = & \Xi_\xi \xi_2 + \Xi_\rho \rho_2 + \Xi_{\xi\xi} \xi_1^2 + 2\Xi_{\xi\rho} \xi_1 \rho_1 + \Xi_{\eta\eta} \eta_1^2 + \Xi_{\rho\rho} \rho_1^2 \\ & + \Xi_{\sigma\sigma} (\rho_1')^2 + \Xi_{\mu\mu} (\phi_1')^2 - \Xi_0 \phi_1^2 - 2H_\eta \eta_1 \phi_1, \end{aligned} \quad (5.34d)$$

$$\begin{aligned} \Lambda_3 = & \Xi_\xi \xi_3 + \Xi_\rho \rho_3 - 3(\Xi_\xi \xi_1 + \Xi_\rho \rho_1) \phi_1^2 \\ & + 3[\Xi_{\xi\xi} \xi_1 \xi_2 + \Xi_{\xi\rho} (\xi_1 \rho_2 + \xi_2 \rho_1) + \Xi_{\eta\eta} \eta_1 \eta_2 + \Xi_{\rho\rho} \rho_1 \rho_2 + \Xi_{\sigma\sigma} \rho_1' \rho_2' + \Xi_{\mu\mu} \phi_1' \phi_2'] \\ & + [\Xi_{\xi\xi\xi} \xi_1^3 + 3\Xi_{\xi\xi\rho} \xi_1^2 \rho_1 + 3\Xi_{\xi\eta\eta} \xi_1 \eta_1^2 + 3\Xi_{\xi\rho\rho} \xi_1 \rho_1^2 + 3\Xi_{\xi\sigma\sigma} \xi_1 (\rho_1')^2 \\ & + 3\Xi_{\xi\mu\mu} \xi_1 (\phi_1')^2 + 3\Xi_{\rho\eta\eta} \rho_1 \eta_1^2 + \Xi_{\rho\rho\rho} \rho_1^3 + 3\Xi_{\rho\sigma\sigma} \rho_1 (\rho_1')^2 + 3\Xi_{\rho\mu\mu} \rho_1 (\phi_1')^2] \\ & - 3\Xi_0 \phi_1 \phi_2 - 3H_\eta (\eta_1 \phi_2 + \eta_2 \phi_1) \\ & - 3[H_{\xi\eta} \xi_1 \eta_1 + H_{\rho\eta} \xi_1 \eta_1 + H_{\sigma\mu} \rho_1' \phi_1'] \phi_1. \end{aligned} \quad (5.34e)$$

We now summarize our results for the generic cases (m, Y) and $(X, \bar{n})$. In Case $(0, Y)$, we have

$$(1 + \gamma_0) = \Xi_0/H_\eta, \quad \gamma_0 \neq 0, \quad (5.35a)$$

$$\xi = \gamma_0 + (\gamma_2 - H_\eta/\Xi_0)\alpha^2/2 + 0(\alpha^4), \quad (5.35b)$$

$$\rho = \bar{\rho}(\gamma_0) - [P_\xi(\gamma_2 - H_\eta/\Xi_0) - P_{\eta\eta}]\alpha^2/2P_\rho + 0(\alpha^4), \quad (5.35c)$$

$$\eta = \alpha + 0(\alpha^3), \quad (5.35d)$$

$$\phi = -H_\eta\alpha/\Xi_0 + 0(\alpha^3), \quad (5.35e)$$

$$\gamma = \gamma_0 + \gamma_2\alpha^2/2 + 0(\alpha^4), \quad (5.35f)$$

$$\Lambda = \Xi_0 + [(\Xi_\xi P_\rho - P_\xi \Xi_\rho)(\gamma_2 - H_\eta/\Xi_0)/P_\rho + \Xi_{\eta\eta} + H_\eta^2/\Xi_0]\alpha^2/2 + 0(\alpha^4). \quad (5.35g)$$

That the errors in (5.35b,c,f,g) are $0(\alpha^4)$ follows easily from (5.31), which implies that $\xi_3 = 0$, $\rho_3 = 0$, $\gamma_3 = 0$.

In Case $(\bar{m}, Y)$, we have

$$p(\gamma_0) = \bar{m}^2 \pi^2, \quad (5.36a)$$

$$\xi = \gamma_0 + \left\{ \gamma_2 + C_2 + \frac{[(P_\rho - 4\bar{m}^2 \pi^2 \Sigma_\sigma) D_2 - \Xi_\rho F_2]}{\Xi_\xi \Sigma_\sigma (q - 4\bar{m}^2 \pi^2)} \cdot \cos 2\bar{m}\pi S \right\} \frac{\alpha^2}{2} + 0(\alpha^4), \quad (5.36b)$$

$$\rho = \bar{\rho}(\gamma_0) + \left\{ -\frac{P_\xi}{P_\rho} (\gamma_2 + C_2) - \frac{E_2}{P_\rho} + \frac{(\Xi_\xi F_2 - P_\xi D_2)}{\Xi_\xi \Sigma_\sigma (q - 4\bar{m}^2 \pi^2)} \cos 2\bar{m}\pi S \right\} \frac{\alpha^2}{2} + 0(\alpha^4), \quad (5.36c)$$

$$\eta = \alpha \cos \bar{m}\pi S + 0(\alpha^3), \quad (5.36d)$$

$$\phi = -(H_\eta \alpha / \Xi_0) \cos \bar{m}\pi S + 0(\alpha^3), \quad (5.36e)$$

$$\gamma = \gamma_0 + \gamma_2 \alpha^2 / 2 + 0(\alpha^4), \quad (5.36f)$$

$$\Lambda = \Xi_0 + \left\{ \frac{(\Xi_\xi P_\rho - \Xi_\rho P_\xi)}{P_\rho} (\gamma_2 + C_2) - \frac{\Xi_\rho E_2}{P_\rho} + \frac{[(\Xi_\xi P_\rho - \Xi_\rho P_\xi) - 4\bar{m}^2 \pi^2 \Xi_\xi \Sigma_\sigma]}{\Xi_\xi \Sigma_\sigma (q - 4\bar{m}^2 \pi^2)} + \Xi_{\eta\eta} + \frac{H_\eta^2}{\Xi_0} \right\} \frac{\alpha^2}{2} + 0(\alpha^4). \quad (5.36g)$$

The constants C_2 , D_2 , E_2 , F_2 are given in (5.29).

In Case (X, $\bar{n}$), we have

$$q(\gamma_0) = \bar{n}^2 \pi^2, \quad (5.37a)$$

$$\xi = \gamma_0 - \alpha a (\Xi_\rho / \Xi_\xi) \cos \bar{n}\pi S + \gamma_2 \alpha^2 / 2 + \frac{\alpha^2 a^2 [A_2 - \bar{n}^2 \pi^2 \Xi_{\sigma\sigma}]}{4\Xi_\xi} \left[\frac{P_\xi}{3\bar{n}^2 \pi^2 \Sigma_\sigma \Xi_\xi} - 1 \right] \cos 2\bar{n}\pi S + 0(\alpha^3 a^3), \quad (5.37b)$$

$$\rho = \bar{\rho}(\gamma_0) + \alpha a \cos \bar{n}\pi S - \alpha^2 P_\xi \gamma_2 / 2P_\rho - \alpha^2 a^2 [B_2 + \bar{n}^2 \pi^2 P_{\sigma\sigma}] / 4P_\rho - \frac{\alpha^2 a^2 [B_2 - \bar{n}^2 \pi^2 P_{\sigma\sigma} - (A_2 - \bar{n}^2 \pi^2 \Xi_{\sigma\sigma}) P_\xi / \Xi_\xi]}{12\bar{n}^2 \pi^2 \Sigma_\sigma} \cos 2\bar{n}\pi S + 0(\alpha^3 a^3), \quad (5.37c)$$

$$\eta = 0, \quad (5.37d)$$

$$\phi = 0, \quad (5.37e)$$

$$\gamma = \gamma_0 + \gamma_2 \alpha^2 / 2 + 0(\alpha^3), \quad (5.37f)$$

$$\Lambda = \Xi_0 + \frac{\alpha^2 a^2 [\Xi_{\xi\xi} \Xi_\rho^2 - 2\Xi_{\xi\rho} \Xi_\xi \Xi_\rho + \Xi_{\rho\rho} \Xi_\xi^2]}{2\Xi_\xi^2} + 0(\alpha^3 a^3). \quad (5.37g)$$

The vanishing of η_k and ϕ_k for all k is a consequence of the form of the perturbation equations. A formal proof of this can be obtained by induction.

When the alternative boundary conditions (2.18) are used, we find that the corresponding modification of (5.1) has nontrivial solutions only if γ_0 is a positive solution of

$$p(\gamma_0) = 4\bar{m}^2 \pi^2 \quad \text{or} \quad \cot [p^{1/2}(\gamma_0)/2] = -2p^{1/2}(\gamma_0) M_\mu / (1 + \gamma_0) \Xi_0. \quad (5.38a,b)$$

It is not hard to see that the smallest positive root of (5.38) (if it exists) must be a root of (5.38b) and that this root γ_0 satisfies $p(\gamma_0) \in (\pi^2, 4\pi^2)$. Therefore the first eigenvalue of shear type when (2.18) is used is larger than the first such eigenvalue when (2.17e) is used.

It should be noted that in this case the resulting representations for η and ϕ would show their most pronounced effects at the middle of the beam rather than at the ends. This is consistent with the observed behavior of yield-point effects (cf. [10, pp. 10ff.]), which hold when the supports are better approximated by boundary conditions (2.18) than by (2.17e).

Justification of the perturbation process. When one uses this kind of formal perturbation process, there is the risk that the representations for the bifurcating branches may be valid only for $\alpha = 0$, i.e., bifurcation does not occur and all one's labor is in vain. (Examples of this phenomenon are well-known. See [13] for examples.) Fortunately for the generic cases we have treated in detail, it can be proved (by invoking a version of the implicit function theorem) that there do in fact exist bifurcating branches of solutions with the representations we have provided. (A proof for the pure necking problem that employs the Poincaré shooting method is given in [2].) In cases such as $(0, \bar{n})$ and $(\bar{m}, \bar{n})$, which we have passed over, such proofs become far more difficult (cf. [15], [16]) because the structure of solutions may be far more complicated (cf. [8, 9]). In such circumstances we could also appeal to a variety of abstract theorems that justify the formal perturbation methods under weaker hypotheses on multiplicity and under somewhat stronger hypotheses on material behavior. (The most powerful of these theorems hold for variational problems.)

6. Interpretation of results

In this section we interpret our results by examining how well they could describe the effects observed in real materials, which are usually treated by various models of plasticity. We can visualize the bifurcation processes embodied in (5.35)–(5.37) by Figure 3 in which we sketch the curves generated by giving

$$\nu(\eta) \equiv \left\{ \int_0^1 \eta^2 dS \right\}^{1/2} \text{sign } \eta(0),$$

$$\nu(\rho) \equiv \left\{ \int_0^1 [\rho - \bar{\rho}(\gamma_0)]^2 dS \right\}^{1/2} \text{sign } [\rho(0) - \bar{\rho}(\gamma_0)],$$

and γ as functions of the parameter α . (This figure also serves as a schematic diagram for the bifurcation process in the infinite dimensional space where ξ , ρ , η , ϕ are found.)

We first note that this diagram is symmetric about the $(\gamma, \nu(\rho))$ -plane (as a consequence of the symmetry conditions (2.16)). The trivial solutions of Section 3 are indicated by heavy lines. They include the straight unsheared state represented

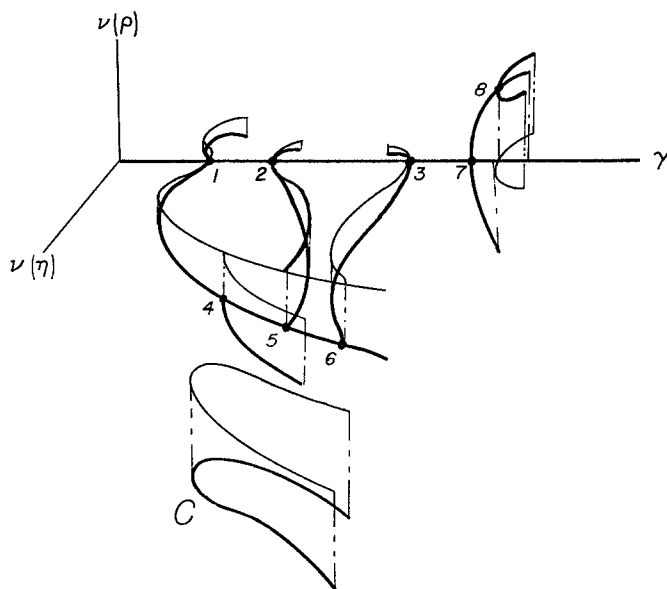


Figure 3.

by the heavy line drawn along the γ -axis as well as the solution branches of case $(0, Y)$ indicated as bifurcating from the points 1 and 2. The representation (5.35) shows that these branches are tangent to the $(\gamma, \nu(\eta))$ -plane at the γ -axis. The very fine curves represent the projections of these branches onto the $(\gamma, \nu(\eta))$ -plane. The trivial, uniformly-sheared solution emanating from point 1 may itself have bifurcation points 4, 5, 6, ... on it. (These bifurcation points may be studied as those in Section 5, but this study is more complicated because the resulting perturbation equations are not uncoupled, in consequence of the reduced effectiveness of the symmetry conditions (2.16).) It may well happen that the uniformly sheared branch emanating from 2 joins the branch emanating from 1 at a point 5. A variant of this possibility is that the branch from 1 simply returns to the γ -axis at some other point of bifurcation. Our analysis cannot determine whether this occurs. It is also conceivable that there are branches C that do not bifurcate from the straight unsheared state.

At point 3, we show the bifurcation of a solution branch of the type $(\bar{m}, Y)$. The representation (5.36) shows that it is also tangent to the $(\gamma, \nu(\eta))$ -plane at the γ -axis. This branch may well link up with a uniformly sheared branch at a point 6 or else return back to the γ -axis at some other point of bifurcation. (Such a possibility is consistent with observed behavior of the propagation of shear instabilities, accompanied by Lüders bands, throughout bars of certain materials under tension. Cf. the discussion in [10, p. 10] and the photos in [14]. These effects in elastic-plastic materials occur in the vicinity of the yield-point(s).)

At point 7, we show a branch of the type $(X, \bar{n})$. Note that it is confined to the

$(\gamma, \nu(\rho))$ -plane and that it is not symmetric about the γ -axis. This corresponds to a solution in which there is a nonuniform change in thickness, i.e., this solution manifests necking.

The branches coming out of the uniformly sheared states have necking and shear effects coupled, as we have mentioned above. These secondary branches may well join other branches at branching points. The local character of these secondary branches can be computed with difficulty but such an analysis would not predict the global behavior.

It is conceivable that any given branch of solutions has secondary branches. E.g., the necking branch emanating from 7 could split at 8 into branches in which the shear is not zero.

When the boundary conditions (2.18) are used, the bifurcation diagram no longer possesses uniformly sheared branches such as that emanating from 1. The general character of the diagram is the same. It is not unlikely that the uniformly sheared states are replaced by non-trivial solutions that are singular perturbations of the uniformly sheared states for large values of γ_0 . The discrepancy between these solutions would be most pronounced in boundary layers at the ends.

These remarks indicate that the structure of solutions can be exceedingly complicated. We shall make a few definitive remarks about the global behavior of the solutions, but there are at present no available analytic methods that could deliver a detailed picture. For specific choices of the constitutive functions, the problem could be attacked by a variety of effective (but expensive) numerical methods. (Cf. [8] for such a study of shell buckling.) But such analyses cannot distinguish between what is common to all such problems and what is particular to a special problem.

We note that the ordering, number, and very existence of the two different types of eigenvalues differ with different materials. This is a virtue of our model because some materials are not observed to suffer yield-point effects, some materials are not observed to suffer necking, and some materials are not observed to suffer either effect. The solutions of necking type can be sketched from (5.37). The nature of the coefficients in these representations depends upon the material. By varying these coefficients within their admissible ranges, we find that the resulting shapes conform to various kinds of necking (illustrated in [14], e.g. Cf. [2.4] for further details). Solutions of the sort (5.36) do not conform to the observed yield-point effects because the boundary condition (2.17e) is not a good approximant of the actual boundary conditions in the testing apparatus. This defect would be remedied by using the more realistic (but mathematically more difficult) condition (2.18).

To interpret the condition $p(\gamma_0) = m^2 \pi^2$ that signals the onset of yielding, we let $1 + \varepsilon$ be the axial stretch and β be the shear angle defined by

$$\begin{aligned} 1 + \varepsilon &= (1 + \varepsilon) \cos \beta, \\ \eta &= (1 + \varepsilon) \sin \beta. \end{aligned} \tag{6.1}$$

(Cf. (2.5).) We set

$$Q(\varepsilon, \beta, \rho, \sigma, \mu) = (\mathbf{k} \times \mathbf{r}'/|\mathbf{r}'|) \cdot (\Xi(\mathbf{w})\mathbf{a} + H(\mathbf{w})\mathbf{b}) = H(\mathbf{w}) \cos \beta - \Xi(\mathbf{w}) \sin \beta. \tag{6.2}$$

The value of Q is just the component of resultant force in the direction normal to the axis. From (6.2) we obtain

$$Q_\beta \equiv \frac{\partial}{\partial \beta} Q(\gamma_0, 0, \bar{\rho}(\gamma_0), 0, 0) = (1 + \gamma_0)H_\eta - \Xi_0. \quad (6.3)$$

At $\gamma_0 = 0$, $Q_\beta > 0$. As γ_0 is increased Q_β may fall to zero, in which case (5.3) implies that $p(\gamma_0) = 0$. Thus the onset of yielding occurs when Q ceases to be a strictly increasing function of β in the trivial state. For $p(\gamma_0)$ to equal $\bar{m}^2 \pi^2$, Q_β must descend to a certain negative value, namely $-\bar{m}^2 \pi^2 H_\eta M_\mu / \Xi_0$.

It is interesting to note that our constitutive restriction (2.14) ensures the monotonicity of H and not Q . (The physical significance of this is discussed in [1, 3, 5].) Thus Q_β is free to assume negative values.

We can study the condition $q(\gamma_0) = \bar{n}^2 \pi^2$ in a similar way. We find

$$\frac{d}{d\gamma_0} \Xi(\gamma_0, 0, \bar{\rho}(\gamma_0), 0, 0) = (\Xi_\xi P_\rho - \Xi_\rho P_\xi) / P_\rho, \quad (6.4)$$

which we take to be positive at $\gamma_0 = 0$. γ_0 is an eigenvalue of necking type when (6.4) falls to one of the negative values $-\bar{n}^2 \pi^2 \Sigma_\sigma \Xi_\xi / P_\rho$. Thus necking does not occur until Ξ_0 passes a local maximum and has started to decrease, in conformity with evidence for real materials. It is important to note that these conditions for the onset of necking branches are completely consistent with the requirement that $\partial \Xi / \partial \xi > 0$ (a consequence of (2.14a)). This requirement is suspended in many models of plasticity designed to describe necking.

It is illuminating to study the curves of Λ vs. γ . This is the information recorded from the tension test, the most primitive of all experiments for the determination of material behavior. If α^2 is eliminated from (5.35f,g), (5.36f,g), (5.37f,g), we obtain

$$\Lambda = \Xi_0 + L(\gamma - \gamma_0) / \gamma_2 + O(\alpha^3), \quad (6.5)$$

where L respectively stands for the coefficients of $\alpha^2/2$ in (5.35g), (5.36g), (5.37g).

In Figure 4, we illustrate (6.5). The heavy curve represents Ξ_0 . Points 1 and 2

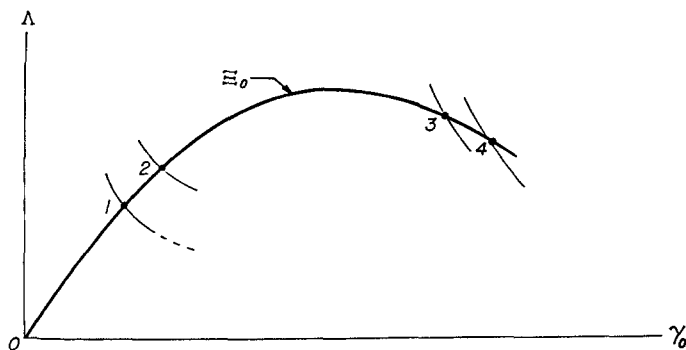


Figure 4.

represent bifurcation points of type $(\bar{m}, Y)$ and points 3 and 4 represent bifurcation points of type $(X, \bar{n})$. Of course, there may be more or fewer such points with any ordering. In the cases we show, $L/\gamma_0 < 0$ and $L/\gamma_0 < d\Xi_0/d\gamma_0$. If the branch emanating from 1 is stable, then as γ_0 is increased beyond its value at 1, Λ will show a drop in value while the bar develops some shearing. Exactly the same phenomenon is observed at the upper yield-point in real materials. Our local analysis cannot say whether or not this bifurcation branch rejoins the main branch for larger values of γ_0 . A similar pronounced drop in the value of Λ also occurs at the points 3 and 4. Such a reduction is manifested in observed necking processes. (Cf. [2, 4] for a fuller discussion of the behavior of necking branches.)

To obtain some global results and to gain some insight into the question of stability, we restrict our attention to hyperelastic rods for which there is a scalar-valued strain-energy function W depending on $\mathbf{w}$ such that the constitutive equations are given by

$$\begin{aligned} \Xi &= \partial W / \partial \xi, & H &= \partial W / \partial \eta, \\ M &= \partial W / \partial \mu, & P + \mu M / (1 + \rho) &= W_\rho. \end{aligned} \quad (6.6)$$

(Cf. [1].)

When (6.6) holds, our boundary value problem consists of the Euler equations and natural boundary conditions for the variational problem of minimizing

$$\int_0^1 W(\mathbf{w}(S)) dS \quad (6.7)$$

subject to

$$\mathbf{r}(1) - \mathbf{r}(0) = \int_0^1 [(1 + \xi)\mathbf{a} + \eta\mathbf{b}] dS = (1 + \gamma)\mathbf{i}.$$

When W is subjected to some mild growth conditions, the results of [1, 7] show that this variational problem has a smooth absolute minimizer in the class of functions for which (6.7) is finite. To show that this minimizer is not the trivial solution we need only show that the second variation of (6.7) at the trivial solution is not positive-definite. The second variation at the trivial solution is

$$\begin{aligned} \int_0^1 \{ [W_{\xi\xi}(\delta\xi)^2 + 2W_{\xi\rho}\delta\xi\delta\rho + W_{\rho\rho}(\delta\rho)^2 + W_{\eta\eta}(\delta\eta)^2 + W_{\sigma\sigma}(\delta\rho')^2 \\ + W_{\mu\mu}(\delta\phi')^2] + \Xi_0[2\delta\eta\delta\phi + (1 + \gamma_0)(\delta\phi)^2] \} dS, \end{aligned} \quad (6.9)$$

where $\delta\xi$, $\delta\eta$, $\delta\rho$, $\delta\phi$ are arbitrary functions (in appropriate spaces) satisfying

$$\int_0^1 \delta\xi dS = 0, \quad \int_0^1 [\delta\eta + (1 + \gamma_0)\delta\phi] dS = 0. \quad (6.10)$$

If we choose

$$\delta\xi = 0, \quad \delta\rho = 0, \quad \delta\eta = \cos m\pi S, \quad \delta\phi = (1 + \gamma_0)^{-1} \cos m\pi S, \quad (6.11)$$

then (6.10) is satisfied and (6.9) is reduced to a positive multiple of

$$(m\pi)^2 - \frac{(1 + \gamma_0)H_\eta}{\Xi_0} p(\gamma_0) \quad (6.12)$$

and if we choose

$$\delta\xi = -(\Xi_\rho/\Xi_\varepsilon) \cos \bar{n}\pi S, \quad \delta\rho = \cos \bar{n}\pi S, \quad \delta\eta = 0, \quad \delta\phi = 0 \quad (6.13)$$

(cf. (5.37)), then (6.10) is satisfied and (6.9) is reduced to a positive multiple of

$$(\bar{n}\pi)^2 - q(\gamma_0). \quad (6.14)$$

Expression (6.9) fails to be positive definite if m and γ_0 are such that (6.12) is negative or if n and γ_0 are such that (6.14) is negative. In light of this, we can summarize our preceding remarks in a Global Non-uniqueness Theorem: *Let W satisfy (2.14) and (2.15) and have suitable growth properties. Let γ_0 and Ξ_0 be positive. If either $p(\gamma_0) > 0$ or $q(\gamma_0) > \pi^2$, there exists a classical nontrivial solution of the boundary value problem that minimizes (6.7) subject to (6.8).* Note that (6.2) and (6.4) imply that the positivity of $p(\gamma_0)$ and $q(\gamma_0) - \pi^2$ are respectively equivalent to the requirements that $Q_\beta < 0$ and $d\Xi_0/d\gamma_0 < -\pi^2 \Sigma_\sigma \Xi_\varepsilon / P_\rho$. A related condition would replace $p(\gamma_0) > 0$ when (2.18) is used. It is customary to interpret these results as showing that the trivial solution is *unstable* and the nontrivial minimizer is *stable* when either $p(\gamma_0) \geq 0$ or $q(\gamma_0) \geq \pi^2$. (There are however serious technical difficulties in proving that these solutions are dynamically stable equilibrium states for suitable dynamical problems when the concept of stability must be both precise and useful.)

Note that a similar analysis can be performed on nonhyperelastic rods by means of degree theory, but the results are not quite so sharp.

Finally we note that our solutions have the capacity to exhibit hysteresis. Let $\bar{\gamma}$ denote the first eigenvalue. The results just stated indicate that the trivial solution is unstable for γ in some interval to the right of $\bar{\gamma}$. The local behavior of the branch bifurcating from $\bar{\gamma}$ is largely determined by the sign of γ_2 , which depends on the first three derivatives of the constitutive functions. (Note that (2.14) only controls the first derivatives of these functions.) In Figure 5, we sketch various possible bifurcation diagrams for the branch emanating from $\bar{\gamma}$ in the hyperelastic case. The ordinate represents some component of $\mathbf{w}$. Branches corresponding to local minimizers of (6.9) are drawn in heavy lines. Their disposition is consistent with the consequences of variational methods. These diagrams, by themselves, cannot describe the temporal evolution of our mechanical system since they merely indicate the dependence of equilibrium solutions of a one-parameter family of problems upon the parameter γ . We can however interpret them by the customary device of regarding them as describing the response of a single "quasistatic" process in which γ is allowed to vary very slowly with time. (Implicit in this approach is an assumption that the dynamical problem possesses a strong dissipative mechanism that ensures that a solution jumping from one static branch can land on another such branch with the vibrations so engendered being merely transient effects.) We consider the process in which γ is slowly increased from Γ_1 to Γ_2 and then slowly

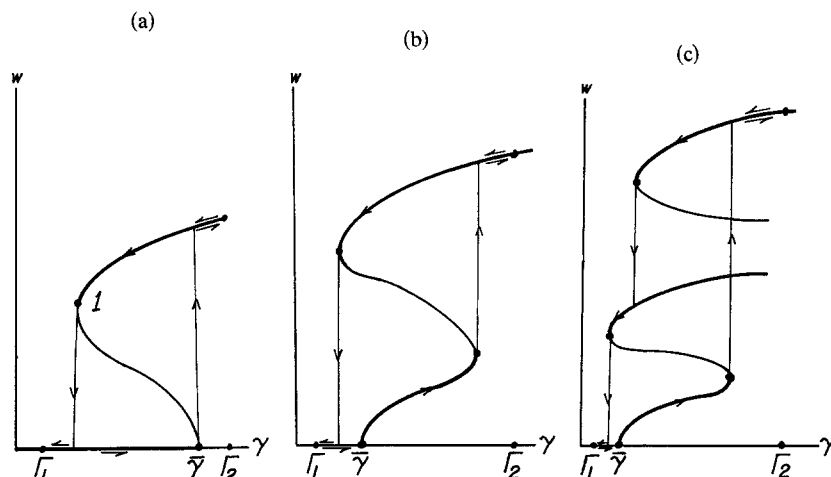


Figure 5.

reduced from Γ_2 to Γ_1 . In Figure 5a, we show the case in which $\gamma_2 < 0$ at $\bar{\gamma}$. The conventional interpretation of the global non-uniqueness result indicates that the trivial solution is unstable for γ in an interval to the right of $\bar{\gamma}$ so that we expect the “stable” solution to jump to another branch. Here we show the jump going to a continuation of the branch emanating from $\bar{\gamma}$. When γ is reduced from Γ_2 to Γ_1 , the stable solution stays on the nontrivial branch until point 1 is reached, whereupon it jumps to the trivial branch. Figure 5b differs from Figure 5a only in that γ_2 is taken positive in the former. In Figure 5c, we show the complexity of response that may well occur when there are a multitude of branches. There is no reason for a solution to jump to a continuation of its original branch as in Figures 5a, b. Moreover, when there are several available stable branches at a given value of γ , there is no static criterion to identify which branch a jumping solution will land upon. (The dynamical criteria employ the notion of domain of attraction in the function space in which dynamical solutions are found.)

7. Conclusion

When we applied our convergent perturbation scheme to the quasilinear eighth-order system of ordinary differential equations (2.2)–(2.5), (2.8)–(2.11), we reduced it to a sequence of *third-order, linear* systems of ordinary differential equations. This uncoupling, a consequence of the symmetry conditions (2.16), made this perturbation scheme practicable. We were even able to use some of the results of the perturbation analysis in the study of global multiplicity and stability of solution.

Throughout our analysis we tacitly exploited the monotonicity condition to prevent the vanishing of various denominators. This condition also plays a fundamental role in the existence theory underlying our global multiplicity result.

We found that the solutions of our boundary value problem for different elastic materials satisfying the monotonicity condition could exhibit a remarkably complex response including shear instabilities, necking instabilities, and hysteresis. In Section 6, we interpreted these effects as being qualitatively the same as those exhibited by real materials in the tension test. These effects are usually explained by various models of plasticity. But we emphasize that our results hold for geometrically exact problems of nonlinear elasticity and are due to the nonlinear interactions. In particular, our materials have $\partial\Xi/\partial\xi$ strictly positive and have no memory, in contrast to the models of plasticity used to explain these phenomena.

8. Appendix—The symmetry condition

Since it is not immediately obvious that the symmetry conditions, which are central to our analysis, must have just the form indicated in (2.16), we show how they may be obtained from a two- (or three-) dimensional theory of nonlinear elasticity. For simplicity, we consider the plane strain of a body that in its reference configuration occupies the region

$$0 \leq X^1 \leq 1, \quad -h \leq X^2 \leq h, \quad -\infty \leq X^3 \leq \infty, \quad (8.1)$$

where $\mathbf{X} = (X^1, X^2, X^3)$ are the Cartesian coordinates corresponding to the orthonormal basis $(\mathbf{i}, \mathbf{j}, \mathbf{k})$. We set

$$X^1 = S, \quad X^2 = Y. \quad (8.2)$$

The requirement of plane strain in the (X^1, X^2) -plane means that the position $\mathbf{z}(X^1, X^2, X^3)$ of particle (X^1, X^2, X^3) has the form

$$\mathbf{z}(\mathbf{X}) = \boldsymbol{\pi}(S, Y) + X^3 \mathbf{k}, \quad \boldsymbol{\pi} \cdot \mathbf{k} = 0. \quad (8.3)$$

We generate a one-dimensional theory by assuming that $\boldsymbol{\pi}$ is so constrained that

$$\boldsymbol{\pi}(S, Y) = \mathbf{r}(S) + \theta(Y)\mathbf{p}(S), \quad (8.4)$$

where θ is a given odd function of Y . We assume that

$$\left[\frac{\partial \mathbf{z}}{\partial X^1} \times \frac{\partial \mathbf{z}}{\partial X^2} \right] \cdot \frac{\partial \mathbf{z}}{\partial X^3} > 0.$$

Evaluating the left side of this expression at $Y=0$, we get (2.7). We set

$$\mathbf{a}(S) = -\mathbf{k} \times \mathbf{p}(S)/|\mathbf{p}(S)|, \quad \mathbf{b}(S) = \mathbf{p}(S)/|\mathbf{p}(S)|. \quad (8.5)$$

The equilibrium equations for a body subject to zero body force are

$$\frac{\partial}{\partial X^K} \boldsymbol{\tau}^K(\mathbf{X}) = \mathbf{0}, \quad \frac{\partial \mathbf{z}}{\partial X^K} \times \boldsymbol{\tau}^K = \mathbf{0}, \quad (8.6)$$

where $\{\boldsymbol{\tau}^K\}$ are the Piola-Kirchhoff traction vectors. We employ the summation convention in (8.6). If we assume that the tractions on the material surfaces $Y = \pm h$ are zero, then we immediately find from (8.6) that

$$\frac{d}{dS} \int_{-h}^h \boldsymbol{\tau}^1 dY = \mathbf{0}, \quad (8.7)$$

$$\frac{d}{dS} \int_{-h}^h \theta(Y) \boldsymbol{\tau}^1 dY - \int_{-h}^h \theta'(Y) \boldsymbol{\tau}^2 dY + \int_{-h}^h \theta(Y) \frac{\partial \boldsymbol{\tau}^3}{\partial X^3} dY = \mathbf{0}, \quad (8.8)$$

$$\frac{d}{dS} \int_{-h}^h \mathbf{z} \times \boldsymbol{\tau}^1 dY + \int_{-h}^h \frac{\partial}{\partial X^3} (\mathbf{z} \times \boldsymbol{\tau}^3) dY = \mathbf{0}. \quad (8.9)$$

A material is *elastic* if

$$\boldsymbol{\tau}^K(\mathbf{X}) = T^{KL}(C_{PQ}(\mathbf{X}), \mathbf{X}) \frac{\partial \mathbf{z}}{\partial X^L}(\mathbf{X}), \quad (8.10)$$

where

$$C_{PQ} = \frac{\partial \mathbf{z}}{\partial X^P} \cdot \frac{\partial \mathbf{z}}{\partial X^Q}. \quad (8.11)$$

($\{C_{PQ}\}$ are the components of the Green deformation tensor and $\{T^{KL}\}$ are the components of the second Piola-Kirchhoff stress tensor.) If we now substitute the plane strain hypothesis (8.3) into (8.11) and (8.10) and then substitute (8.10) into (8.8) and (8.9), we find that these equations reduce to

$$\frac{d}{dS} \int_{-h}^h \theta \boldsymbol{\tau}^1 dY - \int_{-h}^h \theta'(Y) \boldsymbol{\tau}^2 dY = \mathbf{0}, \quad (8.12)$$

$$\frac{d}{dS} \int_{-h}^h \mathbf{k} \cdot (\mathbf{z} \times \boldsymbol{\tau}^1) dY = 0. \quad (8.13)$$

The substitution of the constraint (8.4) and the equilibrium equation (8.7) in (8.13) reduces (8.13) to

$$\frac{d}{dS} \left[\mathbf{k} \cdot \left(\mathbf{p} \times \int_{-h}^h \theta \boldsymbol{\tau}^1 dY \right) \right] + \mathbf{k} \cdot \left[\mathbf{r}' \times \int \boldsymbol{\tau}^1 dY \right] = 0. \quad (8.14)$$

If we identify

$$\hat{\Xi} \mathbf{a} + \hat{H} \mathbf{b} = \int_{-h}^h \boldsymbol{\tau}^1 dY, \quad (8.15)$$

$$\hat{\Sigma} = \mathbf{b} \cdot \int_{-h}^h \theta \boldsymbol{\tau}^1 dY, \quad \hat{\mathbf{P}} = \mathbf{b} \cdot \int_{-h}^h \theta'(Y) \boldsymbol{\tau}^2 dY, \quad (8.16)$$

$$\hat{M} = \mathbf{k} \cdot \left[\mathbf{p} \times \int_{-h}^h \theta \boldsymbol{\tau}^1 dY \right], \quad (8.17)$$

and if we use (2.2)–(2.5), then we find that (8.7), (8.12), and (8.14) reduce to the equilibrium equations (2.8)–(2.11).

We obtain constitutive equations (2.13) for these variables by substituting (8.3), (8.4), (8.11), (8.12) into (8.15)–(8.17):

$$\begin{aligned} \hat{\Xi}(S) \equiv \Xi(\mathbf{w}(S)) \equiv \mathbf{a} \cdot \int_{-h}^h \{ T^{11}(C_{PQ}(\mathbf{X}), \mathbf{X})[\mathbf{r}'(S) + \theta(Y)\mathbf{p}'(S)] \\ + T^{12}(C_{PQ}(\mathbf{X}), \mathbf{X})\theta'(Y)\mathbf{p}(S) \} dS, \quad \text{etc.}, \end{aligned} \quad (8.18)$$

where in terms of the variables of (2.2)–(2.6)

$$C_{11} = (1 + \xi)^2 + \eta^2 + 2\theta(Y)[\sigma\eta - (1 + \xi)(1 + \rho)\mu] + \theta^2(Y)[(1 + \rho)^2\mu^2 + \sigma^2], \quad (8.19a)$$

$$C_{12} = \theta'(Y)(1 + \rho)[\eta + \theta(Y)\sigma], \quad (8.19b)$$

$$C_{22} = [\theta'(Y)]^2(1 + \rho)^2, \quad (8.19c)$$

$$C_{31} = 0, \quad C_{32} = 0, \quad C_{33} = 1. \quad (8.19d)$$

Note that C_{22} is invariant under sign changes of η , σ , μ , that C_{12} changes sign as η , $\sigma \rightarrow -\eta$, $-\sigma$, and that C_{11} is unchanged as η , $\sigma \rightarrow -\eta$, $-\sigma$.

Let us assume that our material is homogeneous so that $\{T^{KL}\}$ do not depend explicitly on $\mathbf{X}$. Furthermore, let us assume that T^{11} and T^{22} are even in C_{12} and that $T^{12} = T^{21}$ is odd in C_{12} for fixed values of the other arguments. From (8.18) and related formulas we can then read off that Ξ , P , and M are invariant and H and Σ change sign as η , $\sigma \rightarrow -\eta$, $-\sigma$. This accounts for some of the symmetries listed in (2.16). The proofs of the remaining symmetries rely on the oddness of θ and are much subtler. We illustrate one case of such a demonstration. We readily find that

$$\begin{aligned} M(\xi, \eta, \rho, \sigma, \mu) = - \int_{-h}^h (1 + \rho)\theta(Y)[(1 + \xi) + \theta(Y)(1 + \rho)\mu] \\ \cdot T^{11}(C_{11}(S, Y), C_{12}(S, Y), C_{22}(S, Y), 0, 0, 1) dY. \end{aligned} \quad (8.20)$$

Then (8.19) and the oddness of θ imply that

$$\begin{aligned} M(\xi, \eta, \rho, -\sigma, -\mu) = \int_{-h}^h (1 + \rho)\theta(-Y)[(1 + \xi) + \theta(-Y)(1 + \rho)\mu] \\ \cdot T^{11}(C_{11}(S, -Y), C_{12}(S, -Y), C_{22}(S, -Y), 0, 0, 1) dY. \end{aligned} \quad (8.21)$$

If we now change the variable of integration from Y to $Z = -Y$ we find by comparing the resulting form of (8.21) with (8.20) that

$$M(\xi, \eta, \rho, -\sigma, -\mu) = -M(\xi, \eta, \rho, \sigma, \mu), \quad (8.22)$$

which is one of the symmetries of (2.16).

We finally note that the monotonicity conditions (2.14) are consequences of the strong ellipticity conditions for the three-dimensional theory. The proof of this fact relies on the representation of the form (8.18). A simple demonstration of this under the most general of circumstances is given in [6].

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